مقدمة شبكات

المحاضرة الخامسة

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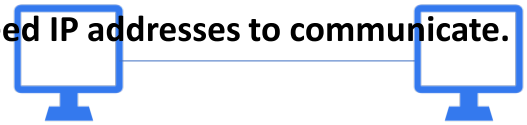
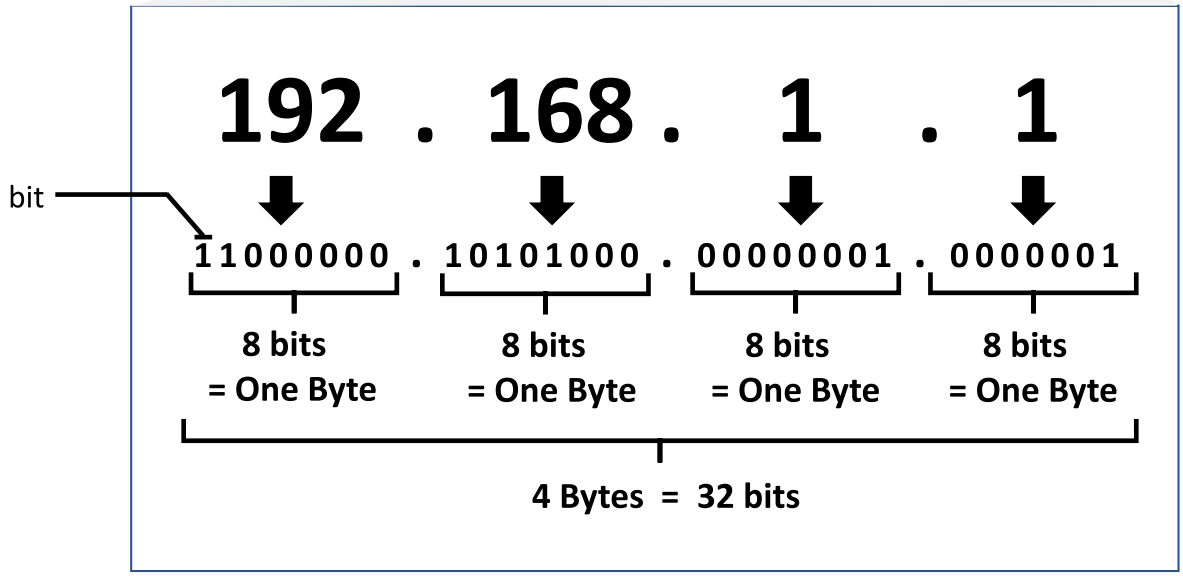
IPv4 Addressing : Introduction

- Computers need IP addresses to communicate.



192 . 168 . 1 . 1 192 . 168 . 1 . 2

- An IP address is structured as follows :



IPv4 Addressing : Numbering Systems

- Humans prefer to use Decimal numbers :
  - Decimal numbers : 1, 2, 3,4, 5, 6, 7, 8, 9, 10, 11, …… , 100,101, ……..

- Computers prefer to use Binary numbers :
  - Binary numbers : 0, 1

- Decimal to Binary conversion :

192

128	64	32	16	8	4	2	1
1	1	0	0	0	0	0	0

Verification : 128 + 64 + 0 + 0 + 0 + 0 + 0 + 0 = 192

IPv4 Addressing : Numbering Systems Exercise

- Convert the following from Decimal to binary:

120 = ……

18 = ……

255 = ……

IPv4 Addressing : Numbering Systems Exercise Solution

120 = 011111000

18 = 00010010

255 = 11111111

IPv4 Addressing : Numbering Systems Exercise

- Convert the following IP address to binary :

192 . 168 . 1 . 1

11000000 . 10101000 . 00000001 . 0000001

IPv4 Addressing : Network Mask

Every computer needs an IP address and mask to communicate. •

Network portion Host portion

IP address : 192 . 168 . 1 . 1

Mask : 255 . 255 . 255 . 0

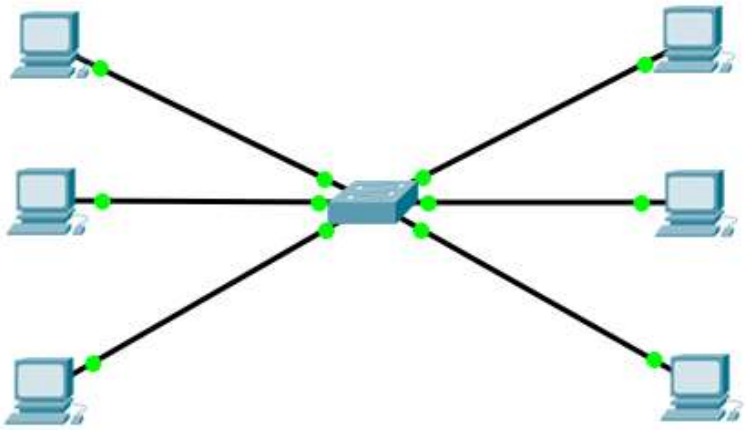
IPv4 Addressing : Network Mask

Computers in this network have the same network portion, so they can communicate. •

192.168. 1 . 1
255.255.255.0

192.168. 1 . 2
255.255.255.0

192.168. 1 . 3
255.255.255.0



192.168. 1 . 4
255.255.255.0

192.168. 1 . 5
255.255.255.0

192.168. 1 . 6
255.255.255.0

IPv4 Addressing : Network Mask

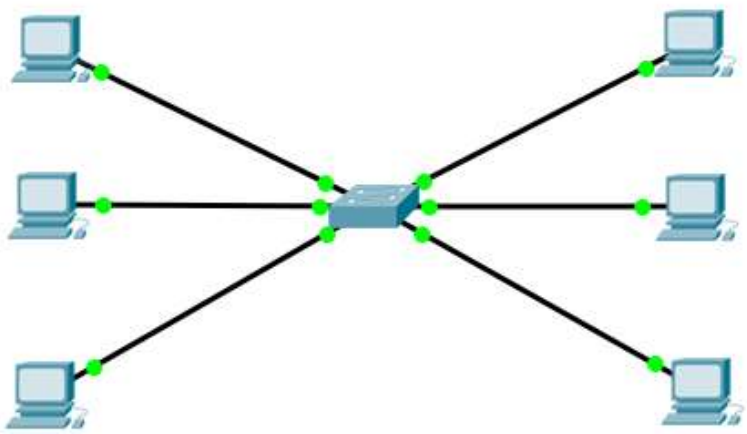
- Computers that do not have the same network portion can not communicate.

- **Check your understanding :**
  - Which PCs can communicate in the following network ?

192.168. 1 . 1
255.255.255.0

192.168. 1 . 2
255.255.255.0

192.168. 2 . 3
255.255.255.0



192.168. 1 . 4
255.255.255.0

192.168. 2 . 5
255.255.255.0

192.168. 1 . 6
255.255.255.0

IPv4 Addressing : IP address and Network Mask

IP addresses and masks can be written either in :

Decimal Notation	or	Binary Notation
192 . 168 . 1 . 1		11000000.10101000.00000001.0000001
255 . 255 . 255 . 0		11111111.11111111.11111111.00000000

Humans prefer to use Decimal

Computers understand Binary

IPv4 Addressing : Address Classes

- IP addresses are categorized into 5 major classes based on the first Byte :

Address Class	Value in First Octet	Classful Mask (dotted decimal)	Classful Mask (prefix notation)
A	1 - 126	255.0.0.0	/8
B	128 - 191	255.255.0.0	/16
C	192 - 223	255.255.255.0	/24
D	224 - 239	N/A	N/A
E	240 - 255	N/A	N/A

- <u>Note</u> : a network mask can be written either in :

Decimal Notation

Example : 255.255.255.0

11111111.11111111.11111111.00000000

CIDR Notation

/24

IPv4 Addressing : Address ranges

- Consider the following IP address :

Network portion | Host portion

192 . 168 . 1 . 1

255 . 255 . 255 . 0

11111111.11111111.11111111.00000000

$$H = 2^n - 2$$
$$= 2^8 - 2$$
$$= 254$$

192 . 168 . 1 . 0

192 . 168 . 1 . 1

Valid Host Range

Number of Hosts :

$H = 254$

192 . 168 . 1 . 254

192 . 168 . 1 . 255

IPv4 Addressing : Classful Addressing Practice

Fill the following table : •

Address	Class	Default Mask	CIDR	Net Portion	Host Portion	Num of Hosts	Host Range	Network Address	Broadcast Address
199.1.4.2	C	255.255.255.0	/24	199.1.4	.2	254	199.1.4.1 – 199.1.4.254	199.1.4.0	199.1.4.255
192.14.2.0	C	255.255.255.0	/24	192.14.2	.2	254	192.14.2.1 - 192.14.2.254	192.14.2.0	192.14.2.255
148.17.9.1	B	255.255.0.0	/16	148.17	.9.1	65534	148.17.0.1 - 148.17.0.254	148.17.0.0	148.17.255.255
193.42.1.1									
126.8.156.0									
220.200.23.1									
200.23.45.58									
177.100.18.4									
119.18.45.0									
29.240.80.7									

IPv4 Addressing : Classful Addressing Practice

Fill the following table : •

Address	Class	Default Mask	CIDR	Net Portion	Host Portion	Num of Hosts	Host Range	Network Address	Broadcast Address
199.1.4.2	C	255.255.255.0	/24	199.1.4	.2	254	199.1.4.1 – 199.1.4.254	199.1.4.0	199.1.4.255
192.14.2.0									
148.17.9.1									
193.42.1.1									
126.8.156.0									
220.200.23.1									
200.23.45.58									
177.100.18.4									
119.18.45.0									
29.240.80.7									

IPv4 Addressing : Subnetting

- We want to subnet the network 192.168.1.0/24 into 2 subnets :

Network portion | Host portion

192 . 168 . 1 . 0

255 . 255 . 255 . 128

11111111.11111111.11111111.10000000

$$H = 2^n - 2$$
$$= 2^7 - 2$$
$$= 126$$

192 . 168 . 1 . 0

192 . 168 . 1 . 1

Valid Host Range

192 . 168 . 1 . 126

Number of Hosts : H = 126

192 . 168 . 1 . 127

192 . 168 . 1 . 128

192 . 168 . 1 . 128

Valid Host Range

192 . 168 . 1 . 254

Number of Hosts : H = 126

192 . 168 . 1 . 255

$N = 2^0 = 1$

/8

11111111	00000000	00000000	00000000
255	0	0	0
10	0	0	0
10	0	0	1
10	255	255	254
10	255	255	255

10.0.0.0

$H = 2^n \quad 2$

$H = 2^{24} \quad 2$

$= 16777214$

$N = 2^0 = 1$ | $N = 2^1 = 2$

/8	/9
11111111 . 00000000 . 00000000 . 00000000	11111111 . 10000000 . 00000000 . 00000000
255 . 0 . 0 . 0	255 . 128 . 0 . 0
	128
10 . 0 . 0 . 0	10 . 0 . 0 . 0
10 . 0 . 0 . 1	10 . 0 . 0 . 1
	$H = 2^{23} \quad 2$
	$= 8388606$
$H = 2^n \quad 2$	
$H = 2^{24} \quad 2$	10 . 127 . 255 . 254
$= 16777214$	10 . 127 . 255 . 255
	10 . 128 . 0 . 0
	10 . 128 . 0 . 1
	$H = 2^{23} \quad 2$
	$= 8388606$
10 . 255 . 255 . 254	10 . 255 . 255 . 254
10 . 255 . 255 . 255	10 . 255 . 255 . 255

10.0.0.0

$N = 2^0 = 1$ | $N = 2^1 = 2$ | $N = 2^2 = 4$

/8	/9	/10
11111111 . 00000000 . 00000000 . 00000000	11111111 . 10000000 . 00000000 . 00000000	11111111 . 11000000 . 00000000 . 00000000
255 . 0 . 0 . 0	255 . 128 . 0 . 0	255 . 192 . 0 . 0
	128	64
10 . 0 . 0 . 0	10 . 0 . 0 . 0	10 . 0 . 0 . 0
10 . 0 . 0 . 1	10 . 0 . 0 . 1	10 . 0 . 0 . 1
		$H = 2^{22} \quad 2$
		$= 4194302$
		10 . 63 . 255 . 254
	$H = 2^{23} \quad 2$	10 . 63 . 255 . 255
	$= 8388606$	10 . 64 . 0 . 0
		10 . 64 . 0 . 1
		$H = 2^{22} \quad 2$
		$= 4194302$
$H = 2^n \quad 2$		
$H = 2^{24} \quad 2$	10 . 127 . 255 . 254	10 . 127 . 255 . 254
$= 16777214$	10 . 127 . 255 . 255	10 . 127 . 255 . 255
	10 . 128 . 0 . 0	10 . 128 . 0 . 0
	10 . 128 . 0 . 1	10 . 128 . 0 . 1
		$H = 2^{22} \quad 2$
		$= 4194302$
		10 . 191 . 255 . 254
	$H = 2^{23} \quad 2$	10 . 191 . 255 . 255
	$= 8388606$	10 . 192 . 0 . 0
		10 . 192 . 0 . 1
		$H = 2^{22} \quad 2$
		$= 4194302$
10 . 255 . 255 . 254	10 . 255 . 255 . 254	10 . 255 . 255 . 254
10 . 255 . 255 . 255	10 . 255 . 255 . 255	10 . 255 . 255 . 255

10.0.0.0

$N = 2^0 = 1$	$N = 2^1 = 2$	$N = 2^2 = 4$	$N = 2^3 = 8$
/8	/9	/10	/11
11111111 . 00000000 . 00000000 . 00000000	11111111 . 10000000 . 00000000 . 00000000	11111111 . 11000000 . 00000000 . 00000000	11111111 . 11100000 . 00000000 . 00000000
255 . 0 . 0 . 0	255 . 128 . 0 . 0	255 . 192 . 0 . 0	255 . 224 . 0 . 0
	128	64	32
10 . 0 . 0 . 0	10 . 0 . 0 . 0	10 . 0 . 0 . 0	10 . 0 . 0 . 0
10 . 0 . 0 . 1	10 . 0 . 0 . 1	10 . 0 . 0 . 1	10 . 0 . 0 . 1
			$H = 2^{21}$ 2
		$H = 2^{22}$ 2	10 . 31 . 255 . 254
		$= 4194302$	10 . 31 . 255 . 255
			10 . 32 . 0 . 0
			10 . 32 . 0 . 1
			$H = 2^{21}$ 2
	$H = 2^{23}$ 2	10 . 63 . 255 . 254	10 . 63 . 255 . 254
	$= 8388606$	10 . 63 . 255 . 255	10 . 63 . 255 . 255
		10 . 64 . 0 . 0	10 . 64 . 0 . 0
		10 . 64 . 0 . 1	10 . 64 . 0 . 1
			$H = 2^{21}$ 2
		$H = 2^{22}$ 2	10 . 95 . 255 . 254
		$= 4194302$	10 . 95 . 255 . 255
			10 . 96 . 0 . 0
			10 . 96 . 0 . 1
			$H = 2^{21}$ 2
$H = 2^n$ 2			
$H = 2^{24}$ 2	10 . 127 . 255 . 254	10 . 127 . 255 . 254	10 . 127 . 255 . 254
$= 16777214$	10 . 127 . 255 . 255	10 . 127 . 255 . 255	10 . 127 . 255 . 255
10.0.0.0	10 . 128 . 0 . 0	10 . 128 . 0 . 0	10 . 128 . 0 . 0
	10 . 128 . 0 . 1	10 . 128 . 0 . 1	10 . 128 . 0 . 1
			$H = 2^{21}$ 2
		$H = 2^{22}$ 2	10 . 159 . 255 . 254
		$= 4194302$	10 . 159 . 255 . 255
			10 . 160 . 0 . 0
			10 . 160 . 0 . 1
			$H = 2^{21}$ 2
	$H = 2^{23}$ 2	10 . 191 . 255 . 254	10 . 191 . 255 . 254
	$= 8388606$	10 . 191 . 255 . 255	10 . 191 . 255 . 255
		10 . 192 . 0 . 0	10 . 192 . 0 . 0
		10 . 192 . 0 . 1	10 . 192 . 0 . 1
			$H = 2^{21}$ 2
		$H = 2^{22}$ 2	10 . 223 . 255 . 254
		$= 4194302$	10 . 223 . 255 . 255
			10 . 224 . 0 . 0
			10 . 224 . 0 . 1
			$H = 2^{21}$ 2
10 . 255 . 255 . 254	10 . 255 . 255 . 254	10 . 255 . 255 . 254	10 . 255 . 255 . 254
10 . 255 . 255 . 255	10 . 255 . 255 . 255	10 . 255 . 255 . 255	10 . 255 . 255 . 255

$N = 2^0 = 1$	$N = 2^1 = 2$	$N = 2^2 = 4$	$N = 2^3 = 8$
/8	/9	/10	/11
11111111 . 00000000 . 00000000 . 00000000	11111111 . 10000000 . 00000000 . 00000000	11111111 . 11000000 . 00000000 . 00000000	11111111 . 11100000 . 00000000 . 00000000
255 . 0 . 0 . 0	255 . 128 . 0 . 0	255 . 192 . 0 . 0	255 . 224 . 0 . 0
	128	64	32
10 . 0 . 0 . 0	10 . 0 . 0 . 0	10 . 0 . 0 . 0	10 . 0 . 0 . 0
10 . 0 . 0 . 1	10 . 0 . 0 . 1	10 . 0 . 0 . 1	10 . 0 . 0 . 1
			$H = 2^{21}$ 2
		$H = 2^{22}$ 2	10 . 31 . 255 . 254
		$= 4194302$	10 . 31 . 255 . 255
			10 . 32 . 0 . 0
			10 . 32 . 0 . 1
			$H = 2^{21}$ 2
	$H = 2^{23}$ 2	10 . 63 . 255 . 254	10 . 63 . 255 . 254
	$= 8388606$	10 . 63 . 255 . 255	10 . 63 . 255 . 255
		10 . 64 . 0 . 0	10 . 64 . 0 . 0
		10 . 64 . 0 . 1	10 . 64 . 0 . 1
			$H = 2^{21}$ 2
		$H = 2^{22}$ 2	10 . 95 . 255 . 254
		$= 4194302$	10 . 95 . 255 . 255
			10 . 96 . 0 . 0
			10 . 96 . 0 . 1
			$H = 2^{21}$ 2
$H = 2^n$ 2			
$H = 2^{24}$ 2	10 . 127 . 255 . 254	10 . 127 . 255 . 254	10 . 127 . 255 . 254
$= 16777214$	10 . 127 . 255 . 255	10 . 127 . 255 . 255	10 . 127 . 255 . 255
10.0.0.0	10 . 128 . 0 . 0	10 . 128 . 0 . 0	10 . 128 . 0 . 0
	10 . 128 . 0 . 1	10 . 128 . 0 . 1	10 . 128 . 0 . 1
			$H = 2^{21}$ 2
		$H = 2^{22}$ 2	10 . 159 . 255 . 254
		$= 4194302$	10 . 159 . 255 . 255
			10 . 160 . 0 . 0
			10 . 160 . 0 . 1
			$H = 2^{21}$ 2
	$H = 2^{23}$ 2	10 . 191 . 255 . 254	10 . 191 . 255 . 254
	$= 8388606$	10 . 191 . 255 . 255	10 . 191 . 255 . 255
		10 . 192 . 0 . 0	10 . 192 . 0 . 0
		10 . 192 . 0 . 1	10 . 192 . 0 . 1
			$H = 2^{21}$ 2
		$H = 2^{22}$ 2	10 . 223 . 255 . 254
		$= 4194302$	10 . 223 . 255 . 255
			10 . 224 . 0 . 0
			10 . 224 . 0 . 1
			$H = 2^{21}$ 2
10 . 255 . 255 . 254	10 . 255 . 255 . 254	10 . 255 . 255 . 254	10 . 255 . 255 . 254
10 . 255 . 255 . 255	10 . 255 . 255 . 255	10 . 255 . 255 . 255	10 . 255 . 255 . 255

$N = 2^0 = 1$

/8

11111111	.	00000000	.	00000000	.	00000000
255	.	0	.	0	.	0

10	**.**	**0**	**.**	**0**	**.**	**0**
10	**.**	**0**	**.**	**0**	**.**	**1**
10	**.**	**255**	**.**	**255**	**.**	**254**
10	**.**	**255**	**.**	**255**	**.**	**255**

10.0.0.0

$H = 2^n \quad 2$

$H = 2^{24} \quad 2$

$= 16777214$